

# DAG Authoring Guide

A practical, end-to-end guide to hand-writing a **workflow DAG** as JSON for the BigBrain workflow engine — defining the tasks, wiring the graph, and passing data between steps without going through the LLM planner.

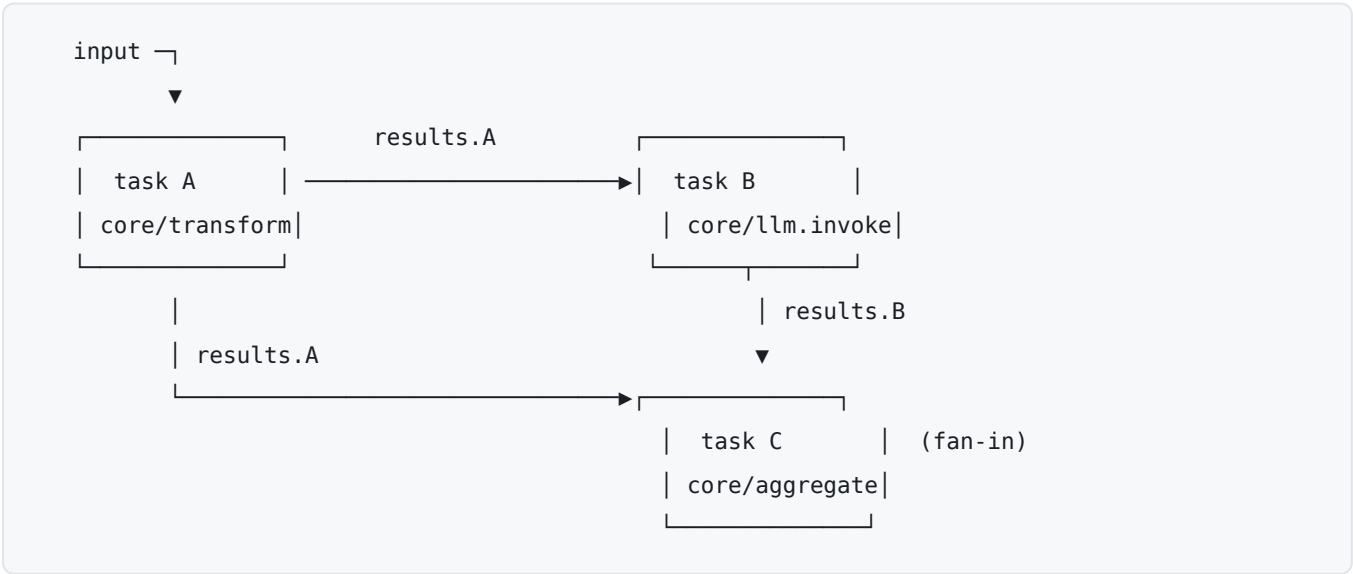
This is the "how do I write the JSON myself" companion to the reference material:

| Doc                                    | What it covers                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>This guide</b>                      | Authoring a workflow definition by hand: shape → tasks → dependencies → data bindings → validation |
| <a href="#">CAPABILITY_CATALOG.md</a>  | The catalog of capability types and their input/output schemas                                     |
| <a href="#">NEURON_ARCHITECTURE.md</a> | How tasks are routed to neurons and executed                                                       |
| <a href="#">WORKFLOW_EVENTS.md</a>     | The lifecycle events a running workflow emits                                                      |

If you want a workflow generated for you from a natural-language goal, use the planner ( `POST /plan` ). Read this guide when you want to author the DAG **directly** — for deterministic pipelines, templates, fixtures, or tests.

## 1. The mental model

A workflow is a **directed acyclic graph (DAG) of tasks**. Each task names a **capability** to run, declares which upstream tasks it **depends on**, and maps data from the shared workflow **context** (and from upstream outputs) into its own input. The engine schedules a task once its dependencies are satisfied, routes it to a handler/neuron by capability type, validates its input and output against the capability's schemas, and writes the result back into the context where downstream tasks can read it.



You author three things: the **tasks**, the **edges** between them ( `dependencies` ), and the **data flow** ( `contextMapping` ). The engine owns everything else.

## 2. The top-level shape

A workflow definition is a JSON object with exactly three required keys:

```
{
  "tasks": { },
  "context": { "input": { }, "results": {}, "shared": {} },
  "artifacts": {}
}
```

| Key       | Type                                  | Meaning                                                                                                         |
|-----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| tasks     | Record<TaskId, Task>                  | The DAG. Keyed by task id; <b>the key must equal the task's own id</b> . At least one task.                     |
| context   | object                                | The shared data space (below). <b>Required</b> — the engine writes task outputs here, so the object must exist. |
| artifacts | Record<ArtifactId, ArtifactReference> | Large/by-reference outputs. <b>Required</b> , usually <code>{}</code> at authoring time.                        |

`context` has three sub-spaces, all required:

```
"context": {
  "input": { "language": "python", "code": "... " },
  "results": {},
  "shared": {}
}
```

- **input** — workflow-level input you provide at creation. This is *your* parameter object; bind into tasks via `input.<field>` .
- **results** — accumulated task outputs, keyed by task id. **Start it empty ( `{}` )**; the engine appends `results.<taskId>` as each task completes.
- **shared** — scratch space for values that aren't a single task's output. Start empty unless you're seeding shared state.

The `tasks` map key and the task's `id` field must match. `"prepare_context": { "id": "prepare_context", ... }` . A mismatch is a definition error.

### 3. Anatomy of a task

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Every task is an object. Here is the full field set; most are optional, and several are runtime fields the engine manages after creation.

```
"prepare_context": {
  "id": "prepare_context",
  "description": "Build the review prompt from user input",
  "requiredCapability": "core/transform",

  "dependencies": [],
  "dependencyCondition": { "type": "all_succeeded" },

  "contextMapping": {
    "include": ["input"],
    "inputMappings": {}
  },
  "input": {
    "expressionType": "jsonata",
    "expression": "{ 'prompt': 'Review this code', 'code': input.code }"
  },

  "status": "pending",
  "attempts": 0,
  "createdAt": "2026-01-01T00:00:00Z",
  "updatedAt": "2026-01-01T00:00:00Z"
}
```

## Fields you author

| Field                                                | Required                                     | Meaning                                                                                                                                                                                            |
|------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>id</code>                                      | ✓                                            | Task identifier. Any <b>non-empty string</b> (the persist schema only requires <code>minLength: 1</code> ). Must equal the <code>tasks</code> map key. See the id-convention note below.           |
| <code>requiredCapability</code>                      | ▲ technically optional, <b>always set it</b> | The capability type that routes this task to a handler, e.g. <code>core/transform</code> , <code>core/llm.invoke</code> , <code>system/bash.exec</code> . Without it the task can't be dispatched. |
| <code>dependencies</code>                            | ✓                                            | Array of upstream task ids. <code>[]</code> = a root task.                                                                                                                                         |
| <code>dependencyCondition</code>                     | optional                                     | When upstreams count as satisfied. Defaults to <code>all_succeeded</code> . See §4.                                                                                                                |
| <code>contextMapping</code>                          | ✓                                            | How data flows into this task. <code>{ include: [...], inputMappings?: {...} }</code> . See §5.                                                                                                    |
| <code>input</code>                                   | optional                                     | Static, literal input for the capability. Merged with the bindings from <code>contextMapping</code> (bindings win on key collision).                                                               |
| <code>description</code>                             | optional                                     | Human-readable purpose. Recommended.                                                                                                                                                               |
| <code>onBehalfOf</code>                              | optional                                     | User id for on-behalf-of (user-scoped) capabilities.                                                                                                                                               |
| <code>inputSchema</code> / <code>outputSchema</code> | optional                                     | JSON Schema snapshotted from the catalog at plan time. You normally omit these by hand; the engine attaches them from the live catalog at dispatch.                                                |
| <code>retry</code>                                   | optional                                     | Opt-in retry policy (below). Absent = fail-fast.                                                                                                                                                   |

## Initial-state fields

The task schema also carries runtime state. At authoring time set them to their initial values:

```
"status": "pending",
"attempts": 0,
"createdAt": "<ISO-8601>",
"updatedAt": "<ISO-8601>"
```

`status` starts at `"pending"`; `attempts` at `0`. `output`, `error`, `leaseId`, `leaseExpiresAt`, `startedAt`, `completedAt` are populated by the engine — leave them out (or `null`). The full status set the engine moves a task through is: `pending` → `ready` → `assigned` → `running` → `completed` (or `failed` / `skipped` / `cancelled`).

## Retry policy (optional)

```
"retry": {
  "maxRetries": 2,
  "backoffMs": 1000,
  "backoffMultiplier": 2,
  "maxBackoffMs": 10000
}
```

Retries are **opt-in per task**. Omit `retry` for fail-fast behaviour. Note the engine only retries failures the handler classifies as retryable.

**Task id convention.** The core persist schema accepts any non-empty string, which is why real workflows use hyphenated ids ( `prepare-context` ) and even ULIDs. **However**, the agentic planner enforces a stricter slug — `^[a-z][a-z0-9_]{0,63}$` (lowercase, digits, underscores; starts with a letter;  $\leq 64$  chars). If your DAG might be extended or patched by the agentic path, use that stricter form (underscores, no hyphens) to stay compatible.

## 4. Wiring the graph

Edges are **explicit** via each task's `dependencies` array — a list of upstream task ids. A task with `"dependencies": []` is a root and runs immediately; a task that lists two upstreams is a fan-in.

```
"aggregate": {
  "id": "aggregate",
  "dependencies": ["llm_anthropic", "llm_openai"],
  "dependencyCondition": { "type": "all_completed" }
}
```

`dependencyCondition.type` controls *when* the dependencies count as satisfied:

| Condition                                        | Fires when...                                                            | Use for                                              |
|--------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|
| <code>all_succeeded</code><br>( <i>default</i> ) | every dependency completed <b>successfully</b>                           | the normal case                                      |
| <code>all_completed</code>                       | every dependency reached a terminal state<br>(success <b>or</b> failure) | fan-in/aggregation that tolerates<br>partial failure |
| <code>any_succeeded</code>                       | at least one dependency succeeded                                        | race / first-good-wins                               |
| <code>any_completed</code>                       | at least one dependency finished                                         | barriers                                             |
| <code>any_failed</code>                          | at least one dependency failed                                           | error-handler / compensation<br>branches             |

**Acyclicity is enforced.** The graph must be a DAG; a cycle is rejected at validation (DFS cycle check). Also: any id you reference in `dependencies` or in a binding must be a task that actually exists in the `tasks` map.

Data bindings imply edges too. If task B binds `results.A.*`, the engine treats B as depending on A even beyond the explicit `dependencies` list — but **author the explicit edge anyway**: it documents intent and keeps the graph readable.

## 5. Passing data between tasks

This is the heart of authoring. A task assembles its input from two places, both inside `contextMapping` :

```
"contextMapping": {
  "include": ["results.prepare_context", "input"],
  "inputMappings": {
    "prompt": "results.prepare_context.prompt",
    "language": "input.language"
  }
}
```

- **include** — the context sub-trees this task is allowed to read. Can be a root ( `"input"` , `"results"` , `"shared"` ) or a narrower path ( `"results.prepare_context"` ). Think of it as the read scope.
- **inputMappings** — the actual wiring: a map of **consumer key** → **context dot-path**. Each entry pulls a value out of the context and places it on the task's input under the consumer key.

### Dot-path roots

A binding's right-hand side is a dot-path rooted at one of exactly three roots:

| Root                                                 | Resolves to               | Example                                    |
|------------------------------------------------------|---------------------------|--------------------------------------------|
| <code>results.&lt;taskId&gt;.&lt;field&gt;...</code> | an upstream task's output | <code>results.llm_anthropic.content</code> |
| <code>input.&lt;field&gt;...</code>                  | workflow-level input      | <code>input.code</code>                    |
| <code>shared.&lt;field&gt;...</code>                 | shared scratch state      | <code>shared.run_id</code>                 |

Any other root is rejected ( `INVALID_CONTEXT_PATH_ROOT` ).

### Nested binding keys

The **consumer key** (left-hand side) may itself be a dot-path, which assembles a nested object on the input:

```
"inputMappings": {
  "value.user": "results.run_whoami.stdout",
  "value.when": "results.run_date.stdout"
}
```

produces the input fragment:

```
{ "value": { "user": "<stdout>", "when": "<stdout>" } }
```

You can also index array positions this way ( `"responses.0"` , `"responses.1"` ) to build an array from multiple upstream outputs.

## Static input + merge rule

`input` holds literal values the capability needs that aren't bound from context:

```
"input": { "model": "claude-sonnet-4-20250514", "maxTokens": 2048 },
"contextMapping": {
  "include": ["results.prepare_context"],
  "inputMappings": { "prompt": "results.prepare_context.prompt" }
}
```

The engine merges the two. **On a key collision, the `contextMapping` binding wins** over the static `input` value. Effective input for the task above: `{ model, maxTokens, prompt }`.

## 6. Capabilities and schemas

`requiredCapability` is a namespaced type — `{owner}/{package}[.{variant}]`, e.g. `core/transform`, `core/llm.invoke`, `system/bash.exec`. It must exist in the **capability catalog**; otherwise validation fails with `CAPABILITY_NOT_IN_CATALOG`.

Each catalog entry carries an **input** and **output** JSON Schema:

```
interface HandlerCatalogEntry {
  type: string; // "core/transform"
  input: JsonSchema; // shape the capability expects
  output: JsonSchema; // shape it returns
}
```

Two things follow for you as an author:

1. **Your task's effective input must satisfy the capability's `input` schema.** Get the field names and types right for the capability you target — they come from the catalog, not from this guide.
2. **Bindings are type-checked.** For every `inputMappings` entry, the engine resolves the producer's output type (from the upstream capability's `output` schema) and confirms it's compatible with the consumer field's type. A mismatch is rejected before the task ever runs.

Schemas are **snapshotted with the workflow** at creation time, so replays use the schemas that were valid when the DAG was authored.

**Sugar primitives are TypeScript-only.** The programmatic builder offers shorthands like `$.bash(...)`, `$.transform(...)`, `$.writeFile(...)`. These do **not** exist in the JSON contract — when hand-writing JSON, always use the longhand capability type and its full `input` object. (For reference, `$.bash` → `system/bash.exec` with `{ command, cwd?, timeoutMs?, env? }`; `$.transform` → `core/transform` with `{ expression, value, expressionType? }`; `$.writeFile` → `mcp/filesystem/write_file` with `{ path, content }`; `$.readFile` → `mcp/filesystem/read_text_file` with `{ path, head?, tail? }`; `$.echo` → `desktop/echo` with `{ message }`.)

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## 7. A complete worked example

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A diamond: prepare a prompt, fan out to two LLMs in parallel, fan back in to aggregate, then synthesize. (Trimmed from `docs/sample-workflow.json`; ids use the planner-safe underscore form.)



```

{
  "tasks": {
    "prepare_context": {
      "id": "prepare_context",
      "description": "Build the code-review prompt from user input",
      "requiredCapability": "core/transform",
      "dependencies": [],
      "dependencyCondition": { "type": "all_succeeded" },
      "contextMapping": { "include": ["input"], "inputMappings": {} },
      "input": {
        "expressionType": "jsonata",
        "expression": "{ 'prompt': 'Review the following code for bugs', 'codeSnippet': input
      },
      "status": "pending", "attempts": 0,
      "createdAt": "2026-01-01T00:00:00Z", "updatedAt": "2026-01-01T00:00:00Z"
    },
    "llm_anthropic": {
      "id": "llm_anthropic",
      "description": "Send the prompt to Claude",
      "requiredCapability": "core/llm.invoke",
      "dependencies": ["prepare_context"],
      "dependencyCondition": { "type": "all_succeeded" },
      "contextMapping": {
        "include": ["results.prepare_context"],
        "inputMappings": { "prompt": "results.prepare_context.prompt" }
      },
      "input": { "model": "claude-sonnet-4-20250514", "promptTemplate": "{{prompt}}", "maxTok
      "retry": { "maxRetries": 2, "backoffMs": 1000, "backoffMultiplier": 2, "maxBackoffMs":
      "status": "pending", "attempts": 0,
      "createdAt": "2026-01-01T00:00:00Z", "updatedAt": "2026-01-01T00:00:00Z"
    },
    "llm_openai": {
      "id": "llm_openai",
      "description": "Send the prompt to GPT-4",
      "requiredCapability": "core/llm.invoke",
      "dependencies": ["prepare_context"],
      "dependencyCondition": { "type": "all_succeeded" },
      "contextMapping": {
        "include": ["results.prepare_context"],
        "inputMappings": { "prompt": "results.prepare_context.prompt" }
      },
      "input": { "model": "gpt-4-turbo", "promptTemplate": "{{prompt}}", "maxTokens": 2048 },
      "status": "pending", "attempts": 0,
      "createdAt": "2026-01-01T00:00:00Z", "updatedAt": "2026-01-01T00:00:00Z"
    },
    "aggregate_responses": {
      "id": "aggregate_responses",
      "description": "Collect both responses into one array",
      "requiredCapability": "core/aggregate",
      "dependencies": ["llm_anthropic", "llm_openai"],
      "dependencyCondition": { "type": "all_completed" },
      "contextMapping": {
        "include": ["results.llm_anthropic", "results.llm_openai"],
        "inputMappings": {
          "responses.0": "results.llm_anthropic.content",

```

```

      "responses.1": "results.llm_openai.content"
    }
  },
  "input": { "strategy": "custom" },
  "status": "pending", "attempts": 0,
  "createdAt": "2026-01-01T00:00:00Z", "updatedAt": "2026-01-01T00:00:00Z"
},

"synthesize": {
  "id": "synthesize",
  "description": "Synthesize a unified report",
  "requiredCapability": "core/llm.invoke",
  "dependencies": ["aggregate_responses"],
  "dependencyCondition": { "type": "all_succeeded" },
  "contextMapping": {
    "include": ["results.aggregate_responses", "results.prepare_context"],
    "inputMappings": {
      "responses": "results.aggregate_responses.responses",
      "language": "results.prepare_context.language"
    }
  },
  "input": { "model": "claude-sonnet-4-20250514", "promptTemplate": "Synthesize these ana"
  "status": "pending", "attempts": 0,
  "createdAt": "2026-01-01T00:00:00Z", "updatedAt": "2026-01-01T00:00:00Z"
}
},

"context": {
  "input": {
    "language": "python",
    "code": "def f(x):\n    return eval(x)"
  },
  "results": {},
  "shared": {}
},
"artifacts": {}
}

```

Walk the graph: `prepare_context` is the root (depends on nothing, reads `input`). `llm_anthropic` and `llm_openai` each depend on it and run in parallel, both binding `results.prepare_context.prompt`. `aggregate_responses` fans in on both with `all_completed` (so it still runs if one model fails), assembling an array via nested keys `responses.0` / `responses.1`. `synthesize` depends on the aggregate and reads back two upstream outputs.

## 8. Validation and common errors

Your definition is validated against the `TypeBox WorkflowDefinitionSchema` on `POST /workflows`, and binding/type checks run when the plan is finalized. What's checked:

- **Shape** — `tasks`, `context` (`input` / `results` / `shared`), and `artifacts` present; each task has its required fields; `id` matches its map key.
- **Acyclicity** — no cycles in the dependency graph.

- **Reference integrity** — every `id` in `dependencies` and every `results.<id>` in a binding names a real task.
- **Capability presence** — every `requiredCapability` exists in the catalog.
- **Binding types** — each `inputMappings` producer type is compatible with the consumer field type.

Error codes you'll see, and what they mean:

| Code                                    | Cause                                                                 | Fix                                                       |
|-----------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|
| <code>CAPABILITY_NOT_IN_CATALOG</code>  | <code>requiredCapability</code> isn't a known type                    | Use a catalog type; check spelling/namespace              |
| <code>INVALID_CONTEXT_PATH_ROOT</code>  | binding path doesn't start with <code>results / input / shared</code> | Re-root the path                                          |
| <code>CONTEXT_PATH_NOT_FOUND</code>     | dot-path doesn't resolve against the producer's output                | Fix the field name / path                                 |
| <code>CONSUMER_INPUT_KEY_UNKNOWN</code> | the capability's input schema has no such field                       | Match the catalog input field names                       |
| <code>TYPE_MISMATCH</code>              | producer output type $\neq$ consumer input type                       | Insert a <code>core/transform</code> , or fix the mapping |

Nested consumer keys ( `value.user` ) skip the consumer-key existence check on the *key* but still type-check the producer side — so verify the capability really accepts a nested object of that shape.

## 9. Parameterizing with templates

There is **no placeholder/variable syntax inside the JSON** (no `${var}` ). DAGs are parameterized through the **context input** plus the template layer:

- **Author for reuse** by reading every variable input from `input.<field>` (bind it through `contextMapping` , never hard-code it in a task's static `input` ).
- **Save a template** ( `POST /workflows/:id/templatize` ): the engine strips runtime state (statuses, attempts, results) and generates an `input_schema` by scanning your `context.input.*` references.
- **Launch from a template** ( `POST /workflows/from-template` ) with an `inputs` object; it's validated against the template's `input_schema` (JSON Schema draft 2020-12), becomes `context.input` , and the DAG runs as normal.

So the path to a parameterized DAG is: bind everything variable from `input.*` → `templatize` → launch with `inputs` .

## 10. Authoring checklist

Before you submit a workflow definition, confirm:

- ☐ **tasks**, **context** ( **input** / **results** / **shared** ), **artifacts** **all present**; **results** and **shared** start as `{}` .
  - ☐ Every **tasks** **key** equals its task's **id** .
  - ☐ Every task sets **requiredCapability** to a real catalog type.
  - ☐ **dependencies** **lists only real task ids**; roots use `[]` .
  - ☐ **No cycles**.
  - ☐ **dependencyCondition** **matches intent** — `all_completed` for fan-ins that tolerate failure; `all_succeeded` otherwise.
  - ☐ Every **binding path** is **rooted** at `results` / `input` / `shared` and resolves to a real field.
  - ☐ **Static** **input** and **bindings** **don't collide** unintentionally (bindings win).
  - ☐ **Variable values** come from `input.*` , not hard-coded, if you'll templatize.
  - ☐ **Initial state set**: `status`: "pending" , `attempts`: 0 , `timestamps` present; no `output` / `error` / `lease` fields.
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## 11. Gotchas

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- **Map key vs id** . They must match exactly. This is the most common hand-authoring mistake.
  - **context is not optional**. Even an all-root workflow needs `"context": { "input": {}, "results": {}, "shared": {} }` — the engine writes results into it.
  - **Don't pre-fill results** . Start it empty; the engine owns it.
  - **Hyphens vs underscores in ids**. Legal for the core engine, but the agentic planner rejects hyphens — prefer underscores for forward-compatibility.
  - **Sugar isn't JSON**. `$.bash(...)` etc. are TypeScript builder helpers; hand JSON uses the longhand `capability` + full input.
  - **Capability input shapes are catalog-defined**. The field names in this guide's examples are illustrative — always check the catalog entry for the exact `input` schema of the capability you target.
  - **Type mismatches fail before execution**. If two steps don't line up, insert a `core/transform` to reshape rather than forcing the binding.
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## Where to go next

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- The capability types and their schemas → [CAPABILITY\\_CATALOG.md](#)
- How a task is routed and executed → [NEURON\\_ARCHITECTURE.md](#)
- A full untrimmed example → [sample-workflow.json](#)
- Lifecycle events to observe a run → [WORKFLOW\\_EVENTS.md](#)